

Short, clear, and job-focused.

TOP MYSQL INTERVIEW QUESTIONS (MOST ASKED)

With explanations + SQL queries

1. Difference between INNER JOIN, LEFT JOIN, RIGHT JOIN

INNER JOIN → Only matching rows

LEFT JOIN → All from left, matching from right

RIGHT JOIN → All from right, matching from left

Query Example:

sql

 Copy code

```
SELECT e.name, d.dept_name
FROM employee e
INNER JOIN department d ON e.dept_id = d.id;
```

2. How to find duplicate records?

sql

 Copy code

```
SELECT name, COUNT(*)
FROM employee
GROUP BY name
HAVING COUNT(*) > 1;
```

3. How to delete duplicate rows (keep only 1)?

sql

 Copy code

```
DELETE e1 FROM employee e1
JOIN employee e2
ON e1.name = e2.name
AND e1.id > e2.id;
```

★ 4. Find 2nd highest salary

Method 1 (using LIMIT):

sql

 Copy code

```
SELECT DISTINCT salary
FROM employee
ORDER BY salary DESC
LIMIT 1 OFFSET 1;
```

Method 2 (subquery):

sql

 Copy code

```
SELECT MAX(salary)
FROM employee
WHERE salary < (SELECT MAX(salary) FROM employee);
```

★ 5. Find Nth highest salary

sql

 Copy code

```
SELECT DISTINCT salary
FROM employee
ORDER BY salary DESC
LIMIT 1 OFFSET N-1;
```

★ 6. Find employees with highest salary in each department

sql

 Copy code

```
SELECT e.*
FROM employee e
JOIN (
    SELECT dept_id, MAX(salary) AS max_salary
    FROM employee
    GROUP BY dept_id
) t
ON e.dept_id = t.dept_id AND e.salary = t.max_salary;
```

★ 7. Count employees in each department

sql

 Copy code

```
SELECT dept_id, COUNT(*)
FROM employee
GROUP BY dept_id;
```

★ 8. Find department with max employees

sql

 Copy code

```
SELECT dept_id, COUNT(*) AS total
FROM employee
GROUP BY dept_id
ORDER BY total DESC
LIMIT 1;
```

★ 9. Find employees who do NOT belong to any department (using LEFT JOIN)

sql

 Copy code

```
SELECT e.*  
FROM employee e  
LEFT JOIN department d ON e.dept_id = d.id  
WHERE d.id IS NULL;
```

★ 10. How to update a specific row?

sql

 Copy code

```
UPDATE employee  
SET salary = 50000  
WHERE id = 101;
```

★ 11. How to delete records?

sql

 Copy code

```
DELETE FROM employee  
WHERE salary < 30000;
```

★ 12. What is a subquery? Example?

sql

 Copy code

```
SELECT name  
FROM employee  
WHERE salary > (  
    SELECT AVG(salary) FROM employee  
);
```

★ 13. How does indexing work?

Index = improves read speed, slows down inserts/updates

Create index:

sql

 Copy code

```
CREATE INDEX idx_emp_name ON employee(name);
```

★ 14. Find employees whose name starts with 'A'

sql

 Copy code

```
SELECT * FROM employee  
WHERE name LIKE 'A%';
```

★ 15. Between vs IN vs EXISTS

Example:

sql

 Copy code

```
SELECT * FROM orders WHERE amount BETWEEN 100 AND 500;
```

```
SELECT * FROM employee WHERE dept_id IN (1, 3, 4);
```

```
SELECT * FROM employee e  
WHERE EXISTS (  
    SELECT 1 FROM department d WHERE e.dept_id = d.id  
);
```

★ 16. Show top 5 salaries

sql

 Copy code

```
SELECT salary FROM employee
ORDER BY salary DESC
LIMIT 5;
```

★ 17. ORDER BY multiple columns

sql

 Copy code

```
SELECT * FROM employee
ORDER BY dept_id ASC, salary DESC;
```

★ 18. HAVING vs WHERE

sql

 Copy code

```
SELECT dept_id, COUNT(*)
FROM employee
GROUP BY dept_id
```